SOFTWARE REQUIREMENTS DOCUMENT

Retail Stock Trading Platform

Full-Stack Web and Mobile Application

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Revision History

Name	Date	Reason For Changes	Version
Mahmoud Elboghdady	May 29, 2026	Initial draft based on stakeholder elicitation meeting	v0.1

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1. Introduction

1.1 Purpose

This Software Requirements Document (SRD) defines the functional and non-functional requirements for a retail stock trading platform commissioned by a Canadian bank. The platform will enable the bank's retail clients to independently execute stock and ETF trades through a web browser and a mobile application, removing the current dependency on in-branch broker interaction. This document is intended to serve as the bridge between the business requirements captured in the Business Requirements Document and the detailed technical specifications that will guide the development team.

1.2 Project Background

The bank currently serves a population of low to medium income retail investors who wish to participate in the stock market. Under the existing process, clients must visit a bank branch and execute trades through a licensed broker using internal bank software. This creates a significant barrier to entry for smaller investors who are deterred by the friction of in-person engagement for routine trades. As a result, the bank faces competitive pressure from self-directed retail trading platforms such as Robinhood, which offer frictionless digital access to the market.

This project aims to eliminate that barrier by delivering a self-service trading platform that empowers clients to manage their own investment accounts. The initial release will be scoped to a Minimum Viable Product (MVP) covering the core trading workflow, with more advanced features planned for future phases.

1.3 Scope

The platform will support the execution of buy and sell orders, as well as basic stop-loss orders, for North American listed stocks and exchange-traded funds (ETFs). Options trading, leveraged products, margin accounts, and foreign exchange trading are explicitly excluded from this phase. The platform will be accessible via a responsive web application and a native mobile application. The expected user base at launch is between 150,000 and 200,000 registered clients.

1.4 Definitions and Abbreviations

Term	Definition
ETF	Exchange Traded Fund, a marketable security tracking an index, commodity, or basket of assets, traded on a stock exchange.
MVP	Minimum Viable Product, the initial release containing only the core features required to deliver value to end users.
TFSA	Tax-Free Savings Account, a registered Canadian account in which investment income and withdrawals are tax-free.

Term	Definition
RRSP	Registered Retirement Savings Plan, a registered Canadian account allowing tax-deferred investment growth.
SRD	Software Requirements Document, this document.
KYC	Know Your Customer, regulatory obligation requiring financial institutions to verify client identity and assess investment suitability.
Atomic Transaction	A database operation that either completes fully or rolls back entirely, ensuring no partial state persisted.
Stop-Loss Order	An instruction to sell a security automatically when its price falls to a specified threshold.

2. System Overview

The Retail Stock Trading Platform is a full-stack application consisting of a web front-end, a mobile application, a back-end API layer, and a persistent database. It interfaces with an external market data provider to retrieve current stock and ETF pricing information and delivers configurable notifications to users via an integrated notification service.

Upon initial setup, a client receives a temporary password issued by the bank following the signing of a trading agreement in-branch. On first login, the user is required to create a new personal password. Subsequent logins are authenticated using the client's bank account number, password, and biometric verification on mobile devices. Once authenticated, clients can view their portfolio across multiple registered account types, search for securities, execute trades, manage a personal watchlist, and configure their notification preferences.

All trade operations are handled as atomic transactions. If any component of the trade execution process fails, including the database write, order routing, or confirmation generation, the entire transaction is rolled back, and the user is notified of the failure. This ensures that no partial or inconsistent trade state ever persisted.

2.1 System Actors

Actor	Description	Primary Interactions
End User (Retail Investor)	A bank client who has signed a trading agreement in-branch and received credentials to access the platform.	Login, portfolio management, stock search, trade execution, watchlist management, notification settings
Market Data API	External third-party service providing North American stock and ETF price data with up to a 5-minute delay.	Supplies real-time price feeds, historical data, and volume information to the platform.
Notification Service	Internal or third-party service responsible for delivering configurable alerts to users.	Sends trade confirmations, price alerts, and volume alerts via configured channels.
Database System	Persistent storage layer maintaining user accounts, trade records, portfolio holdings, and watchlists.	Reads and writes all transactional and user data; enforces atomic trade operations.

3. Functional Requirements

3.1 Authentication and Account Setup

The authentication system governs how users gain and maintain access to the platform. Requirements in this section apply to both the web and mobile applications unless otherwise noted.

FR-AUTH-01 The system shall allow a user to log in using their bank account number and a personal password.

FR-AUTH-02 The system shall enforce a mandatory password reset upon a user's first login when a bank-issued temporary password is detected.

FR-AUTH-03 The system shall support biometric authentication (fingerprint and facial recognition) on mobile devices as an alternative to password entry after the initial login.

FR-AUTH-04 The system shall lock a user account after five consecutive failed login attempts and notify the user via their registered contact method.

FR-AUTH-05 The system shall maintain an authenticated session for a user and shall automatically terminate the session after a configurable period of inactivity.

FR-AUTH-06 The system shall allow a user to log out at any time from any screen within the application.

3.2 Portfolio and Account Management

Each user may hold one or more registered investment account types. The platform must present a unified view while respecting the distinct properties of each account type.

FR-PORT-01 The system shall support three account types per user: Tax-Free Savings Account (TFSA), Registered Retirement Savings Plan (RRSP), and Cash Account.

FR-PORT-02 The system shall display a portfolio summary view showing all account types held by the user, including current total value, individual holdings, and unrealized gain or loss for each position.

FR-PORT-03 The system shall display the available buying power and current cash balance for each account type separately.

FR-PORT-04 The system shall allow the user to navigate between individual account views to inspect holdings and transaction history on a per-account basis.

FR-PORT-05 The system shall display account type labels clearly (TFSA, RRSP, Cash) throughout all relevant screens to prevent user confusion when selecting accounts for trade execution.

> **ASSUMPTION:** Account funding, the mechanism by which a user deposits money into a trading account, has not been confirmed as in scope for this phase. This SRD does not include requirements for linking external bank accounts or initiating deposits. This item is logged as Open Item 2 and must be resolved before development begins.

3.3 Market Data and Stock Discovery

The platform must provide users with sufficient market information to make informed trading decisions. All pricing data is sourced from an external market data API.

FR-MKT-01 The system shall display stock and ETF price data sourced from a third-party market data API, refreshed at a maximum interval of five minutes.

FR-MKT-02 The system shall display an interactive price chart for each security, showing historical price and volume data. The chart must support at a minimum a one-day, one-week, one-month, and one-year view.

FR-MKT-03 The system shall support search for securities by company name or ticker symbol, returning results from North American exchanges (TSX, NYSE, NASDAQ).

FR-MKT-04 The system shall display the following data points for each security: current price, percentage change from previous close, day high, day low, and trading volume.

FR-MKT-05 The system shall clearly indicate when the displayed market data was last refreshed.

FR-MKT-06 The system shall limit the tradeable universe to North American listed stocks and ETFs. Options, leveraged ETFs, foreign exchange, and cryptocurrency instruments shall not be available for trading.

> **ASSUMPTION:** The specific market data API vendor has not been selected. The 5-minute refresh cadence and North American exchange coverage are confirmed requirements. Vendor selection and the resulting integration specification will be addressed during the technical design phase and documented in the Software Requirements Specification.

3.4 Trade Execution

Trade execution is the core workflow of the platform. All trades must be processed as atomic database transactions to ensure data integrity. The user must be able to select which registered account will be used for each trade at the time of order placement.

FR-TRD-01 The system shall allow a user to place a market buy order for any eligible security, specifying the quantity of shares and the account to be debited.

FR-TRD-02 The system shall allow a user to place a market sell order for any security currently held in the selected account, specifying the quantity of shares.

FR-TRD-03 The system shall allow a user to place a basic stop-loss order, specifying a trigger price at which the security will be automatically sold.

FR-TRD-04 The system shall require the user to select a destination account (TFSA, RRSP, or Cash) before confirming any trade.

FR-TRD-05 The system shall display a pre-submission confirmation screen showing the order details (security name, ticker, order type, quantity, estimated price, and selected account), before the order is submitted.

FR-TRD-06 The system shall process all trade operations as atomic transactions. If any step of the trade execution fails, the system shall roll back the entire transaction and present the user with an appropriate error message.

FR-TRD-07 The system shall generate and persist a trade confirmation record upon successful order execution, including: trade ID, timestamp, security name, ticker symbol, order type, quantity, execution price, total value, and account used.

FR-TRD-08 The system shall present the user with a trade confirmation screen upon successful execution, displaying all fields from FR-TRD-07.

FR-TRD-09 The system shall prevent a user from submitting a buy order if the required funds are not available in the selected account.

FR-TRD-10 The system shall prevent a user from submitting a sell order for a quantity exceeding the shares currently held in the selected account.

> **ASSUMPTION:** Trading hours enforcement has not been confirmed. This SRD assumes the system will display a market status indicator (open or closed) and will prevent the immediate execution of market orders outside of market hours. Stop-loss orders are assumed to remain active across sessions until triggered or cancelled. This is logged as Open Item 3 and requires stakeholder confirmation.

3.5 Trade History

Users must be able to review a complete record of their past trading activity, both across all accounts and filtered to a specific account.

FR-HIST-01 The system shall display a chronological trade history for each user, showing all executed orders across all account types.

FR-HIST-02 The system shall allow the user to filter trade history by account type, date range, and security.

FR-HIST-03 The system shall display the following fields for each historical trade: date and time, security name, ticker, order type, quantity, execution price, total value, and account used.

FR-HIST-04 The system shall retain all trade history records for the duration required by applicable Canadian banking and securities regulations.

3.6 Watchlist

Users must be able to monitor securities of interest without holding a position in them. The watchlist serves as a personal tracking tool and as a trigger source for configurable notifications.

FR-WATCH-01 The system shall allow a user to add any eligible security to a personal watchlist.

FR-WATCH-02 The system shall allow a user to remove a security from their watchlist at any time.

FR-WATCH-03 The system shall display a watchlist view showing the current price, percentage change, and volume for each watched security, refreshed in accordance with FR-MKT-01.

FR-WATCH-04 The system shall use the watchlist as the source of securities eligible for price and volume alert notifications, as defined in Section 3.7.

3.7 Notifications

The notification system allows users to receive configurable alerts for trading activity and market events relevant to their watchlist. All notification preferences must be manageable within the application settings.

FR-NOTIF-01 The system shall send a notification to the user upon successful execution of any trade.

FR-NOTIF-02 The system shall send a notification to the user when a security on their watchlist experiences a significant price change, as determined by system-defined thresholds.

FR-NOTIF-03 The system shall send a notification to the user when a security on their watchlist experiences abnormally high trading volume, as determined by system-defined thresholds.

FR-NOTIF-04 The system shall allow each user to independently enable or disable each notification type (trade confirmations, price alerts, volume alerts) through the application settings.

FR-NOTIF-05 The system shall persist user notification preferences and apply them consistently across sessions and devices.

> **ASSUMPTION:** The delivery channel for notifications (push notification, email, SMS, or in-app only) has not been confirmed by the stakeholder. This SRD assumes push notifications are the primary channel on mobile and in-app notifications on web, pending confirmation. This is logged as Open Item 5.

3.8 Quest Page

The elicitation meeting referenced a page labelled 'Quest' as part of the application's navigation. The purpose of this page has not been confirmed. In a regulated financial context, this is likely an investor profile or Know Your Customer (KYC) questionnaire used to assess a client's risk tolerance and investment suitability prior to trading. If this is the case, it would carry regulatory weight and generate a distinct set of requirements in a future revision of this document.

ASSUMPTION: The 'Quest' page is assumed to be an investor profile or risk tolerance questionnaire. No functional requirements have been written for this section pending stakeholder confirmation. This is the highest-priority open item and is logged as Open Item 1.

4. Non-Functional Requirements

4.1 Performance

Performance requirements define the responsiveness and throughput expectations the system must meet under normal and peak operating conditions.

NFR-PERF-01 The system shall complete the submission of a trade order within five seconds of the user confirming the order, measured from the moment the user taps or clicks the confirm button to the moment the confirmation screen is displayed.

NFR-PERF-02 All standard page loads and navigation transitions shall complete within three seconds under normal network conditions.

NFR-PERF-03 The system shall support up to 200,000 concurrent users without degradation of performance below the thresholds defined in NFR-PERF-01 and NFR-PERF-02.

4.2 Availability

Given the nature of investment accounts and user expectations of access to their holdings, the platform must always maintain high availability.

NFR-AVAIL-01 The system shall be available 24 hours a day, 7 days a week, with a target uptime of 99.9% or greater monthly.

NFR-AVAIL-02 Planned maintenance windows shall be scheduled during periods of lowest user activity, and users shall be notified at least 24 hours in advance via in-app message.

NFR-AVAIL-03 The system shall be hosted on a cloud infrastructure platform configured to support automatic failover in the event of a single-node failure, with a recovery time objective not to exceed a few seconds of downtime.

4.3 Security

As a financial application handling sensitive personal and transactional data, security requirements are of the highest priority. The system must implement layered authentication controls and encrypt all data in transit and at rest.

NFR-SEC-01 All data transmitted between the client application and the server shall be encrypted using TLS 1.2 or higher.

NFR-SEC-02 All sensitive data stored in the database, including account credentials, personal information, and trade records, shall be encrypted at rest.

NFR-SEC-03 User passwords shall be stored as salted cryptographic hashes. Plaintext passwords shall never be persisted.

NFR-SEC-04 The system shall require multi-factor authentication for all users. On web, this shall consist of account number and password. On mobile, biometric authentication shall serve as the second factor following initial password login.

NFR-SEC-05 The system shall implement role-based access control to ensure users can only access their own account data. Cross-account access shall be technically prevented at the API layer.

NFR-SEC-06 The system shall log all authentication events, trade submissions, and account modifications with a timestamp and session identifier for audit purposes.

NFR-SEC-07 All temporary passwords issued by the bank shall expire within 72 hours if not used, and shall be invalidated immediately upon first use.

4.4 Scalability

The system must be designed to accommodate growth in the user base beyond the initial 150,000 to 200,000 registered clients without requiring significant architectural changes.

NFR-SCALE-01 The back-end application layer shall be deployed in a horizontally scalable configuration, allowing additional instances to be provisioned dynamically in response to increased load.

NFR-SCALE-02 The database layer shall be architected to support read replicas or equivalent partitioning strategies to handle increased query volume as the user base grows.

4.5 Data Integrity

The integrity of financial transaction data is non-negotiable. The system must ensure that no trade record is lost, corrupted, or inconsistently committed.

NFR-INT-01 All trade operations shall be implemented as atomic database transactions. A trade must either be fully committed or fully rolled back; partial writes are not permitted.

NFR-INT-02 The system shall maintain an immutable audit log of all executed trades. Audit log records shall not be modifiable or deletable by any user, including system administrators.

NFR-INT-03 The system shall retain all trade and account records in compliance with the data retention obligations prescribed by applicable Canadian banking and securities regulations.

4.6 Usability

The platform is designed for retail users who may have limited experience with investment platforms. The interface must be intuitive and accessible to users across a wide range of technical literacy levels.

NFR-USE-01 The application shall be fully functional on the two most recent major versions of Chrome, Safari, Firefox, and Edge on the desktop web.

NFR-USE-02 The mobile application shall be compatible with iOS 15 and above and Android 10 and above.

NFR-USE-03 All critical user actions, login, trade execution, and watchlist management, shall be completable within three taps or clicks from the relevant starting screen.

NFR-USE-04 Error messages presented to the user shall be written in plain language that clearly describes what went wrong and what action the user should take.

5. System Interfaces

5.1 Market Data API

The platform depends on an external market data provider for all pricing and volume information displayed to users and used in trade execution. The specific vendor has not been selected at the time of writing this document; however, the following interface requirements apply regardless of vendor.

IF-API-01 The market data API shall provide price and volume data for all securities listed on the TSX, NYSE, and NASDAQ exchanges.

IF-API-02 The API shall deliver price updates with a maximum delay of five minutes from the last recorded trade on the exchange.

IF-API-03 The API shall support historical price data retrieval sufficient to populate one-day, one-week, one-month, and one-year chart views.

IF-API-04 The system shall handle API unavailability gracefully by displaying a clear message to the user indicating that market data is temporarily unavailable, without crashing or presenting stale data without a timestamp.

5.2 Notification Service

The notification service is responsible for the reliable delivery of alerts to users based on their configured preferences. Whether this is implemented as an internal service or integrated via a third-party provider will be determined during technical design.

IF-NOTIF-01 The notification service shall support delivery of push notifications to mobile devices registered by the user.

IF-NOTIF-02 The notification service shall deliver trade confirmation notifications within 30 seconds of a successful trade execution.

IF-NOTIF-03 The notification service shall deliver price and volume alerts within 10 minutes of the triggering market event.

5.3 Mobile and Web Platforms

The platform must be delivered as both a native mobile application and a browser-based web application. Both interfaces must expose equivalent core functionality, though the interaction patterns and authentication mechanisms may differ as described in the functional requirements.

IF-PLT-01 The web application shall be a responsive single-page application accessible via standard desktop and tablet browsers.

IF-PLT-02 The mobile application shall be a native application available on iOS and Android platforms.

IF-PLT-03 Both interfaces shall consume the same back-end API layer and present consistent data to the user regardless of the access channel used.

6. Data Requirements

This section identifies the principal data entities the system must manage. Detailed data modelling, including entity-relationship diagrams and attribute-level definitions, will be documented in the Software Requirements Specification produced during the technical design phase.

Entity	Key Attributes	Notes
User	User ID, bank account number, hashed password, biometric token reference, registered contact, account creation date	One user may hold multiple account types
Investment Account	Account ID, user ID, account type (TFSA/RRSP/Cash), cash balance, creation date	Each account type tracked separately per user
Security	Ticker symbol, company name, exchange, instrument type (stock/ETF), current price, last updated timestamp	Sourced from market data API; cached locally
Trade Order	Trade ID, user ID, account ID, ticker, order type, quantity, execution price, total value, timestamp, status	Immutable once committed; atomic write required
Watchlist Entry	Entry ID, user ID, ticker symbol, date added	Many-to-many between users and securities
Notification Preference	Preference ID, user ID, notification type, enabled flag, delivery channel	Per user, per notification type
Audit Log	Log ID, user ID, event type, timestamp, session ID, IP address	Append-only; not modifiable post-write

7. Assumptions and Constraints

7.1 Assumptions

The following assumptions have been made in the absence of confirmed stakeholder direction. Each assumption is flagged inline within the relevant requirements section and is also listed here for consolidated review. Any assumption that proves incorrect upon stakeholder review will require a corresponding update to the affected requirements.

ASSUMPTION: Account registration is handled entirely in-branch before first application login. The platform does not support self-service sign-up.

ASSUMPTION: The 'Quest' page referenced during elicitation is an investor profile or KYC risk-tolerance questionnaire. No requirements have been written for this section pending confirmation.

ASSUMPTION: Account funding is out of scope for the initial release. Users are assumed to have pre-funded accounts managed at the banking layer outside this platform.

ASSUMPTION: The system will prevent market order submission outside of exchange trading hours. Stop-loss orders may be placed and modified at any time and will be queued for execution when the market opens.

ASSUMPTION: Push notifications are the primary delivery channel on mobile. In-app notifications serve as the primary channel on the web. Email and SMS are not included in the initial release.

ASSUMPTION: No admin or internal support accounts are required in the first release. Support staff will use separate internal tooling to assist clients.

ASSUMPTION: The platform will comply with all applicable Canadian banking and securities regulations. The specific regulatory obligations will be confirmed and mapped to requirements before development commences.

7.2 Constraints

The following constraints are known limitations or fixed conditions that bound the design and implementation of the system.

CON-01 The tradeable instrument universe is limited to North American listed stocks and ETFs for the initial release. No options, leveraged products, foreign exchange, or cryptocurrency instruments will be supported.

CON-02 Market data will be delayed by up to five minutes. True real-time data feeds are not a requirement of this phase.

CON-03 The platform must be hosted on a cloud provider. On-premises deployment is not in scope.

CON-04 All client-facing text must be presented in English. French language support is not in scope for the initial release.

CON-05 The system must comply with applicable Canadian privacy legislation (PIPEDA) regarding the collection, use, and disclosure of personal information.

8. Open Items

The following items require stakeholder or technical confirmation before the SRD can be considered final. Each item is cross-referenced to the section of this document it impacts. These items should be resolved and incorporated into a revised version of this document prior to the commencement of technical design.

#	Open Item	Impact on SRD	Owner
1	Purpose and content of the 'Quest' page	May introduce KYC/risk-profile requirements and additional functional requirements if regulatory in nature	Stakeholder
2	Account funding mechanism	If in-scope, introduces a separate workflow for linking a bank account and depositing funds	Stakeholder
3	Trading hours enforcement logic	Determines whether limit/stop-loss orders placed outside market hours are queued or rejected	Stakeholder
4	Market data API vendor	Vendor selection may affect data format, latency guarantees, and integration specifications	Technical Lead
5	Notification delivery channels	Confirms whether alerts are delivered via push notification, email, SMS, or a combination	Stakeholder

9. Document Approval

This Software Requirements Document has been prepared by the Business Systems Analyst based on the stakeholder elicitation meeting conducted on May 29, 2026. It is submitted for stakeholder review and approval. By signing below, approvers confirm that the documented requirements accurately reflect the business needs and expectations for the Retail Stock Trading Platform, and acknowledge the open items listed in Section 8 as pending resolution.

Role	Name	Date
Business Owner / DRI		
Project Manager		
Technical Lead		

End of Document